



Sales Forecasting Rossmann Store Sales

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INTRODUCTION



Benefits of Forecasting for Any Business Model

- Forecasting is the process of estimating or predicting future events based on historical data and current analysis. It is a critical element that ensures the sustainability and growth of any project.

Importance of Sales Forecasting for Businesses

- Improved Inventory Management
- Better Financial Planning
- Efficient Workforce Planning
- Smarter Promotion & Marketing Decisions
- Risk Reduction
- Data-Driven Decision Making



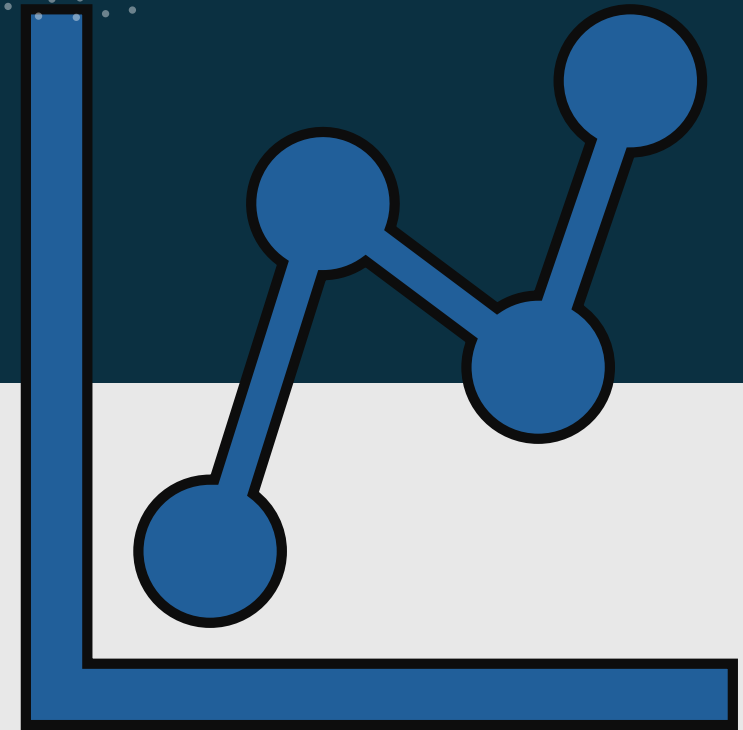
Sales Forecasting Overview

- Sales forecasting for Rossmann stores involves using historical sales data and various factors like promotions, holidays and competitor activity to predict future daily sales, often for up to six weeks in advance.



Scope & Goals

- The dataset contains historical sales data for 1,115 Rossmann stores. The primary objective is to forecast the "Sales" column for the test set.



Data Cleaning

Dataset Overview

- Dataset:
 - Rossmann Sales Forecasting
- Training Data Shape:
 - 1,017,209 rows × 9 columns
- Key Features:
 - Numerical, Categorical, Date
- Store Data Shape:
 - =1119 rows × 10 columns



Data Quality Problems

Large real-world dataset

- requires preprocessing

Issues found after merging with store data:

- Missing values in : (Competition Distance, Competition Open Since Month/Year, Promo2 Since Week/Year, Promo Interval)

Categorical columns not numerical

Different feature scales

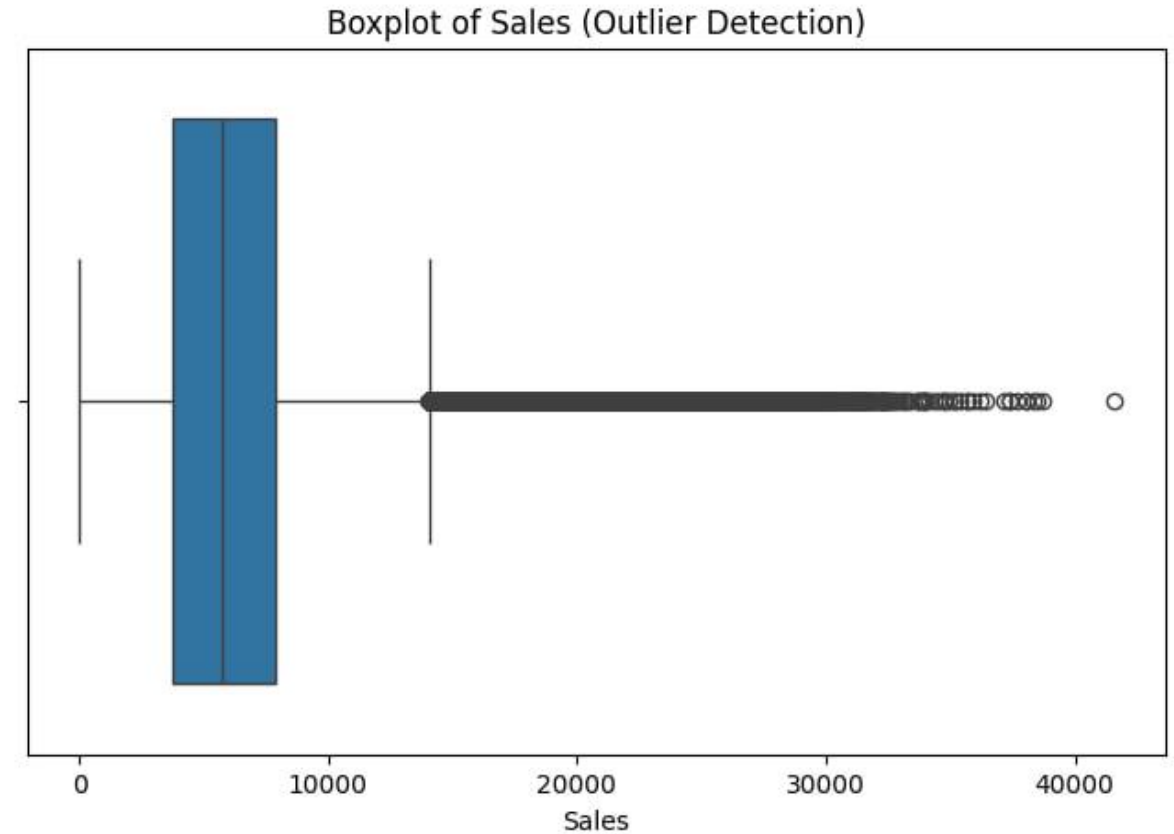
Outliers in Sales

Missing Values & Encoding

- **Missing Values Handling:**
- Categorical (Promo Interval): filled with Most Frequent Value
- Numerical: filled using Iterative Imputer (MICE)
- **Encoding Applied:**
 - Label Encoding:
State Holiday (contained "0", a, b, c) → numeric values
 - One-Hot Encoding :
Promo Interval → Monthly flags
(Promo_Jan, Promo_Feb,... Promo_Dec)
Binary columns created for each month .

Scaling & Outlier Handling

- **Min-Max Scaling:**
 - Normalizes numerical features to same range
 - Prevents dominance of large-scale features (Sales, Customers)
- **Outlier Detection:**
 - Feature: Sales
 - Method: IQR (Interquartile Range)
 - Outliers: 26,694 (2.62%)

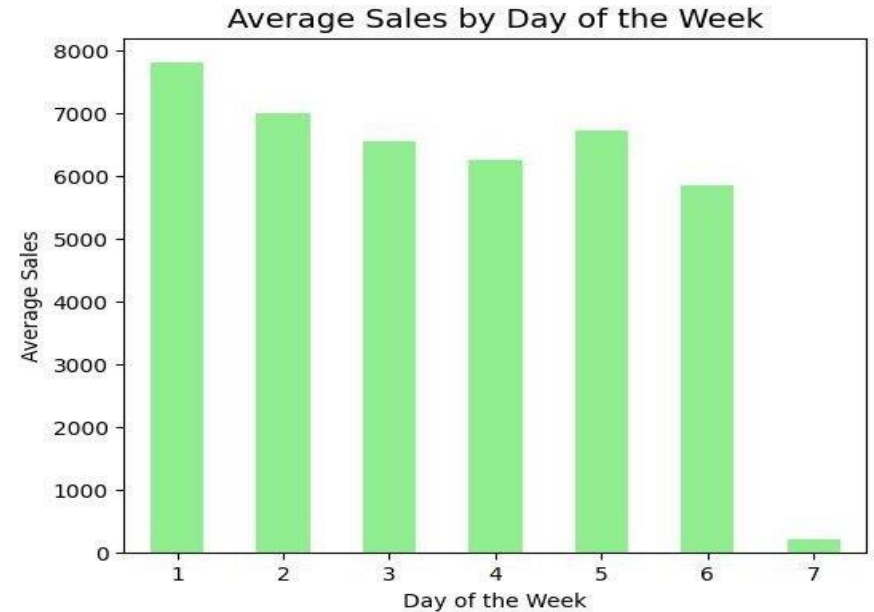
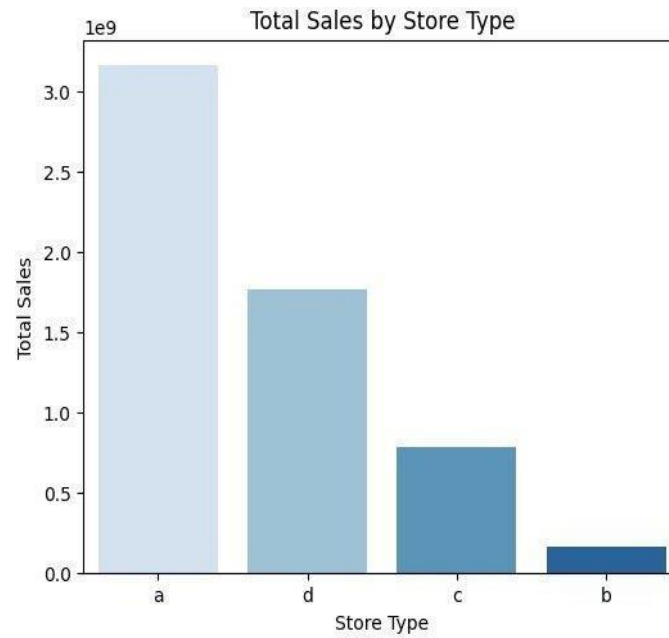
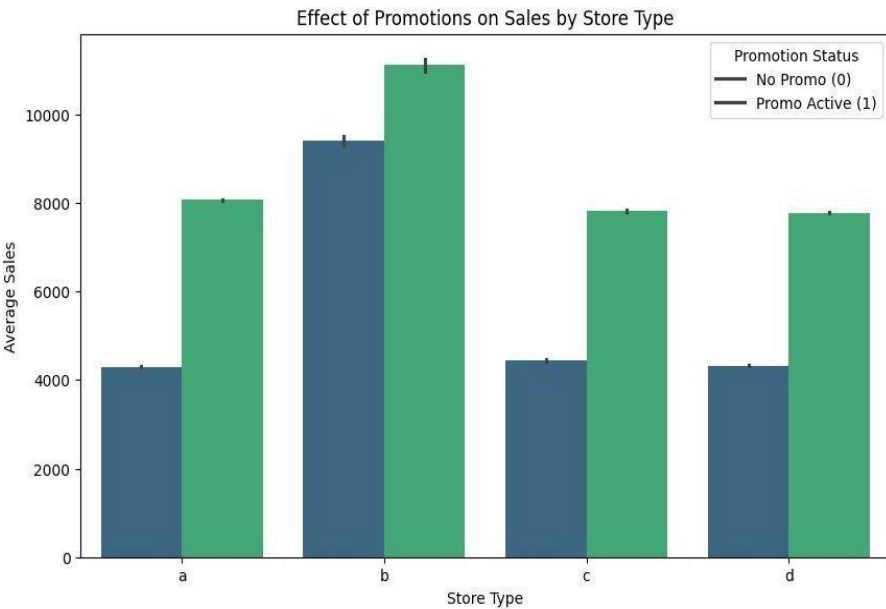
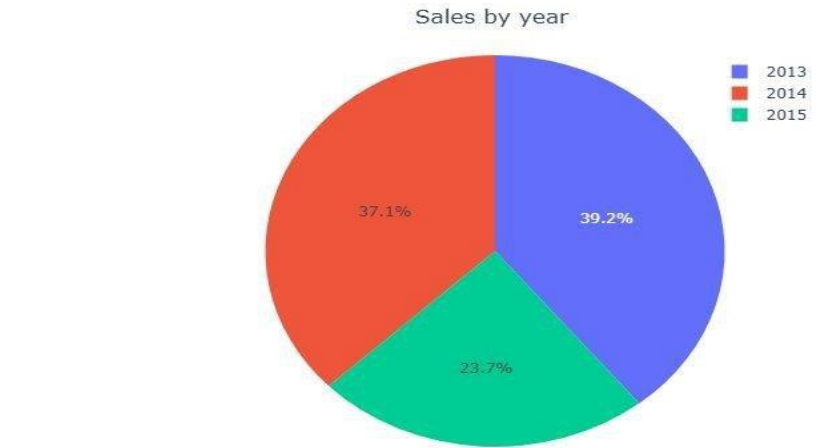
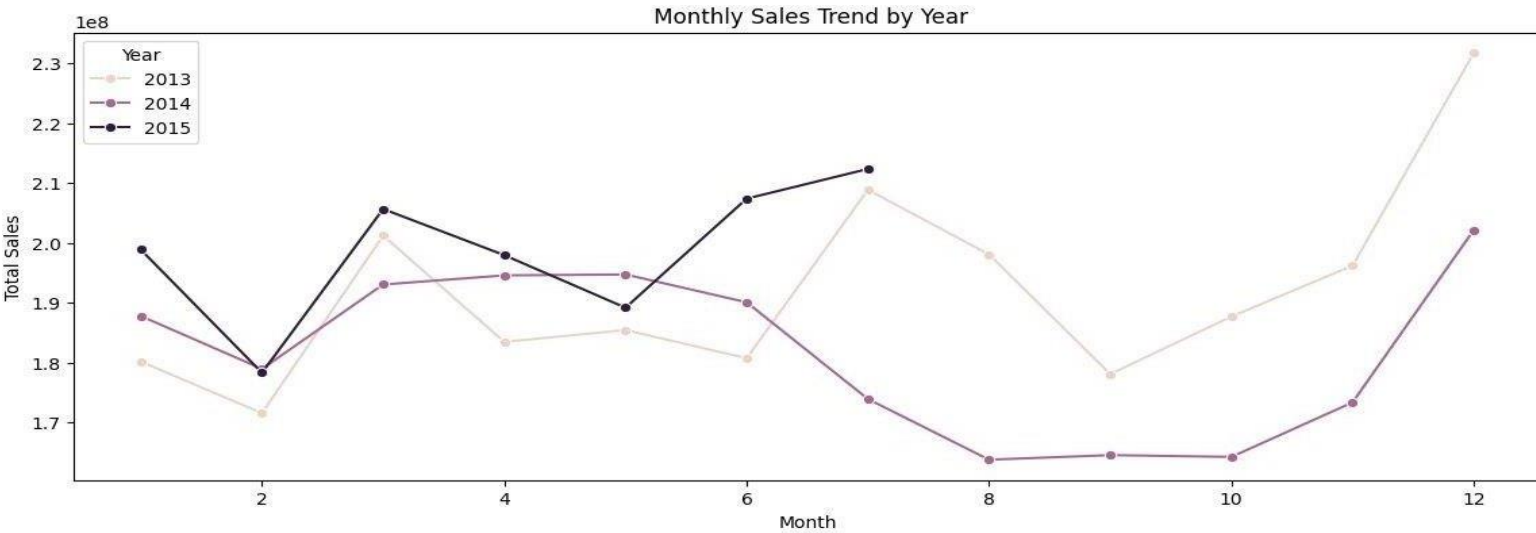


Data Visualization

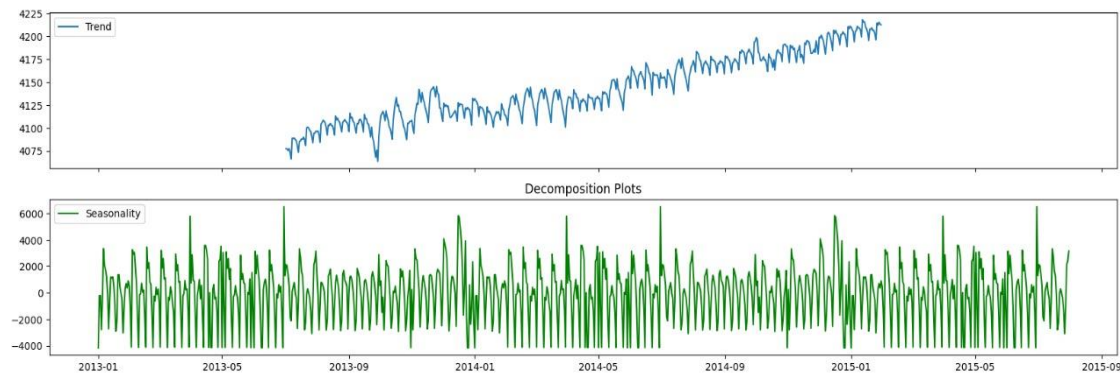
Total Sales
5,873,180,623

Avg Daily Sales
5,774

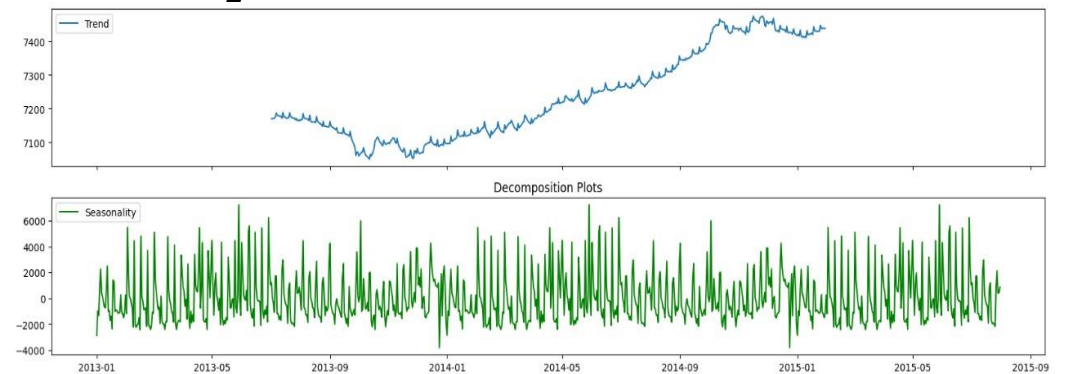
Active Stores
1115



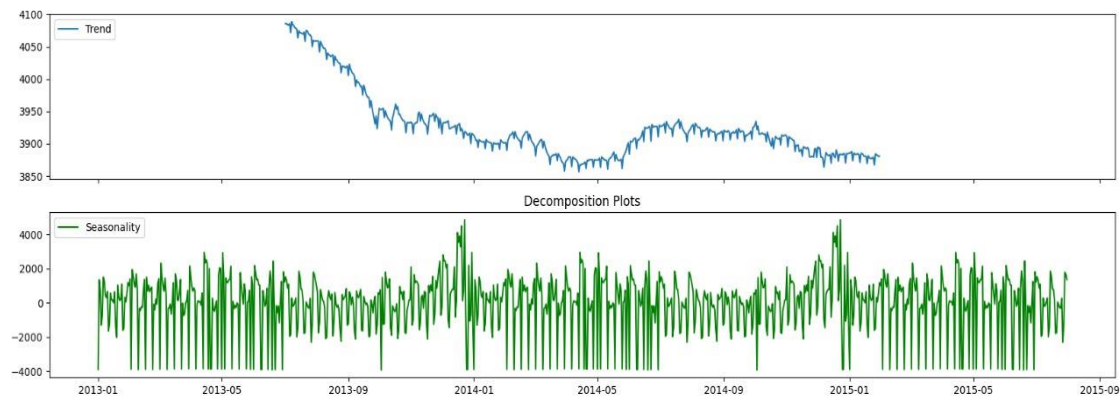
Trend and Seasonality



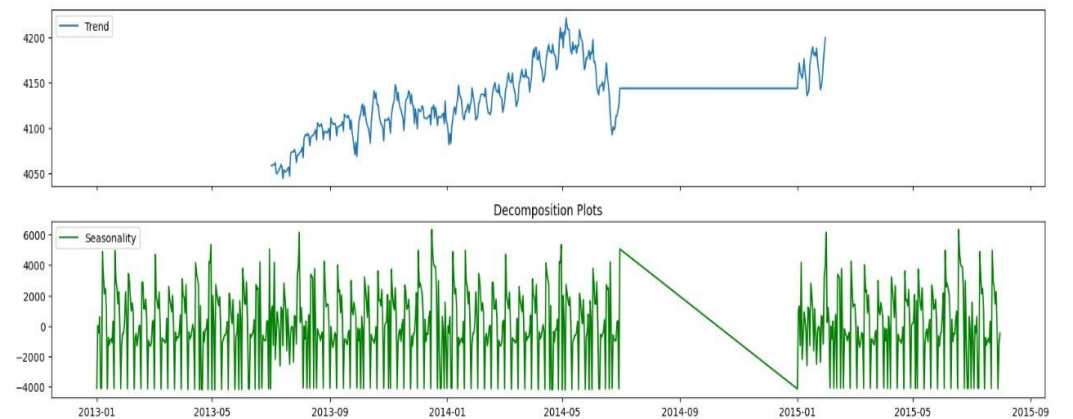
Store a



Store b



Store c



Store d

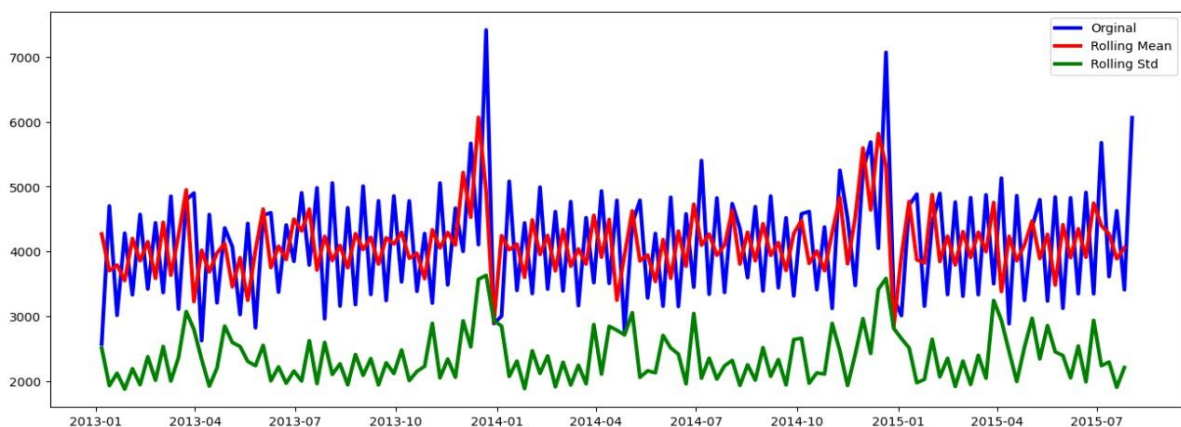
ADF Results & Stationarity Testing

- ADF TEST RESULTS

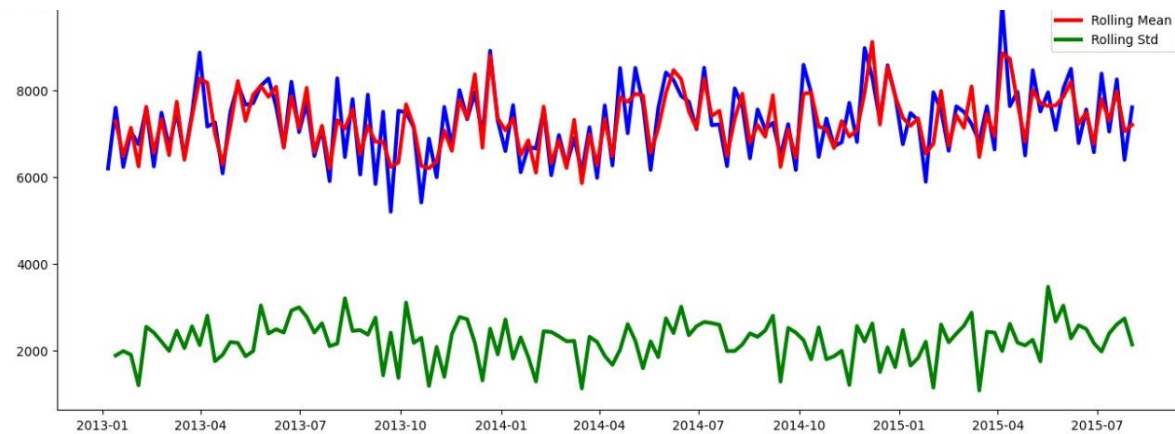
Store	ADF Stat	P-val	Crit 1%	Crit 5%	Crit 10%	Station?
a	-6.218237	0.000000	-3.437477	-2.864686	-2.568445	☒ yes
b	-5.660918	0.000001	-3.437485	-2.864690	-2.568447	☒ yes
c	-4.374784	0.000329	-3.437477	-2.864686	-2.568445	☒ yes
d	-6.237461	0.000000	-3.439253	-2.865469	-2.568862	☒ yes

ADF Results & Stationarity Testing

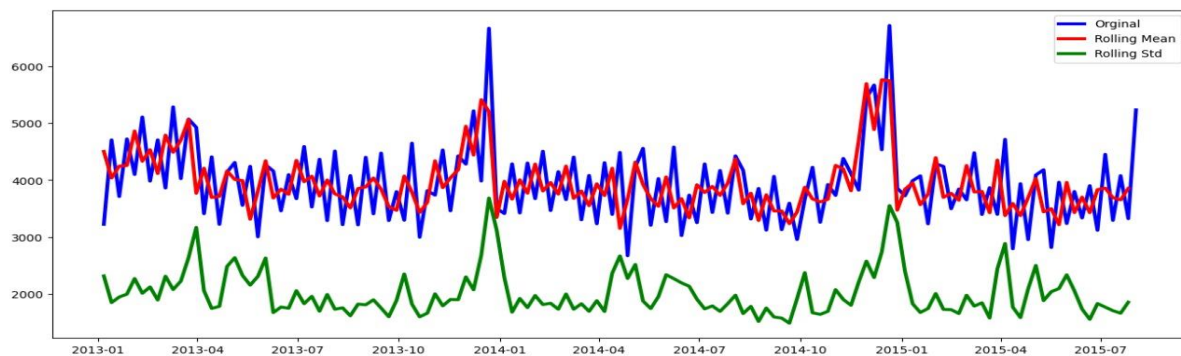
- STATIONARITY PLOTS



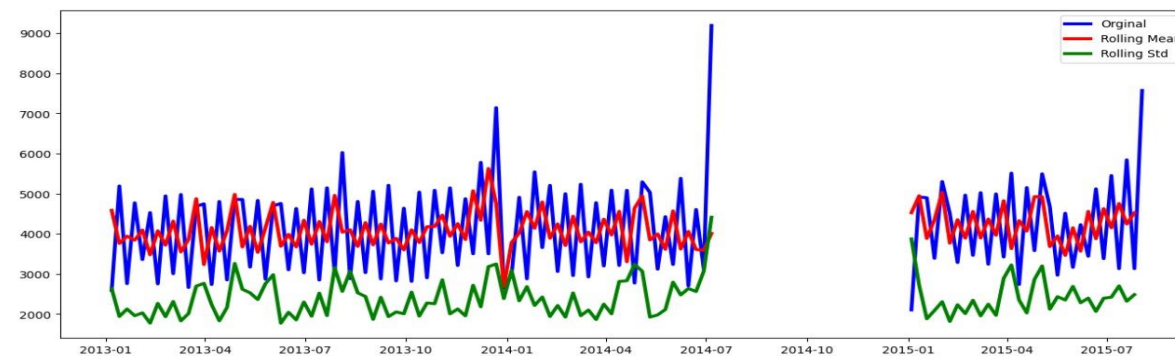
Store a



Store b

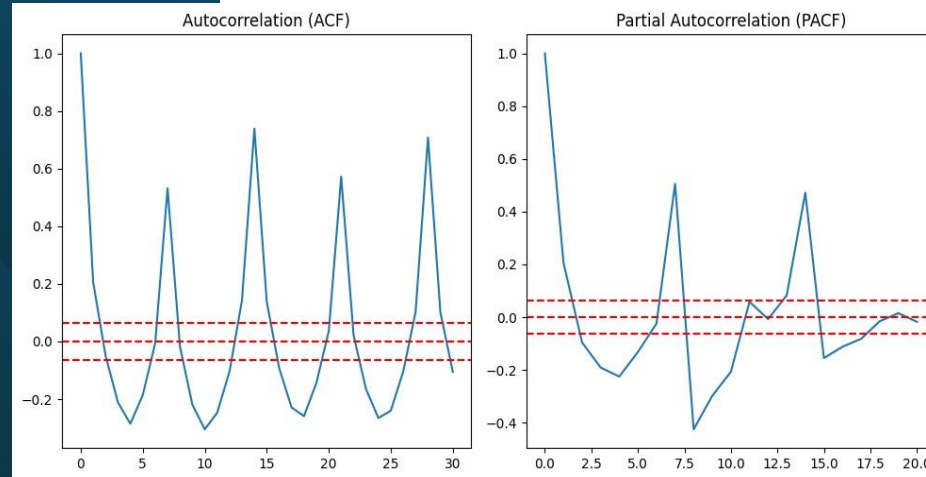


Store c

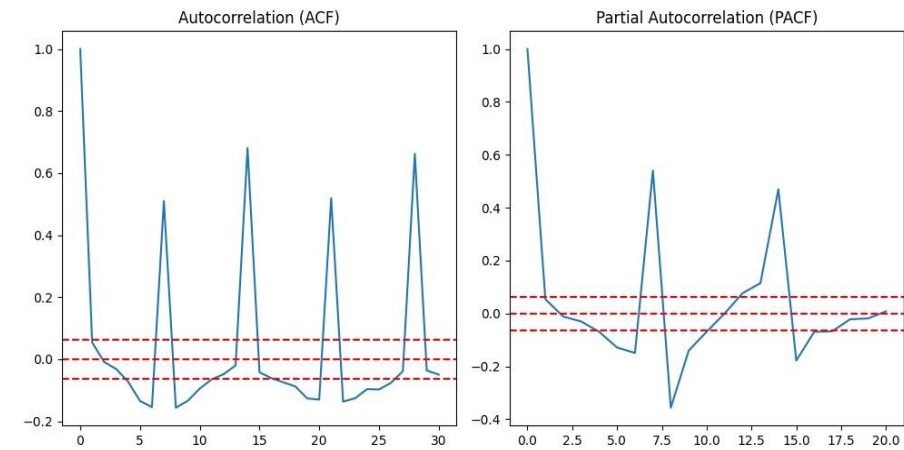


Store d

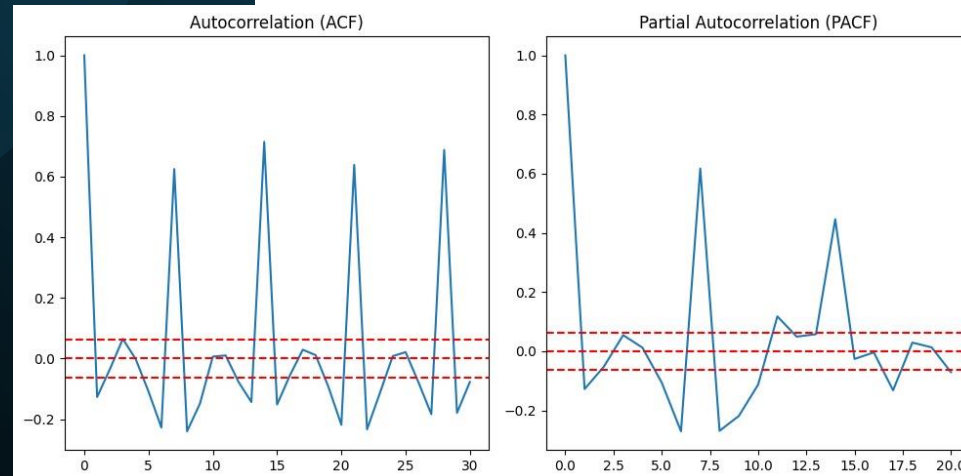
Autocorrelation & Partial Autocorrelation



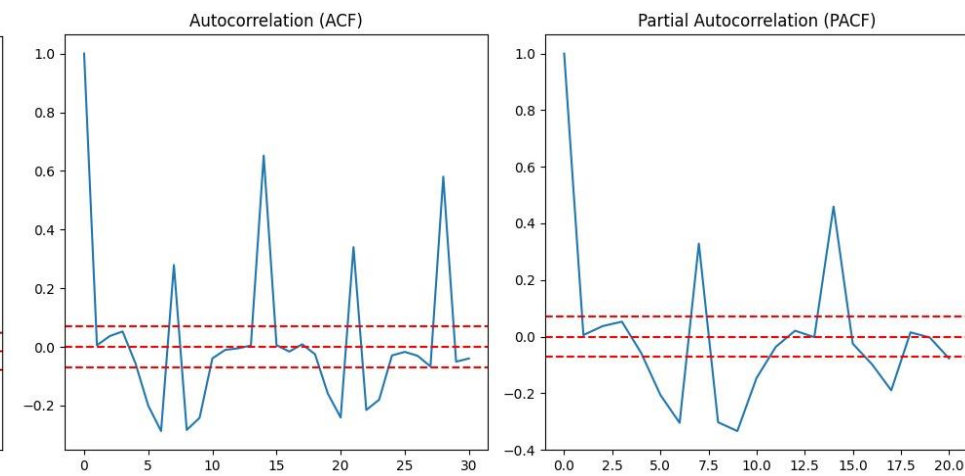
Store a



Store b



Store c

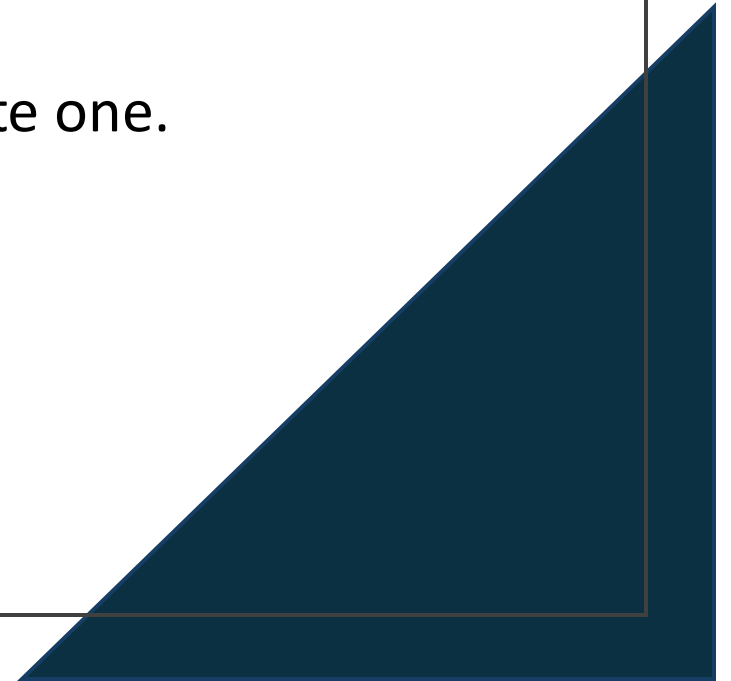


Store d

Modeling Approach

Model Approach Overview

- In our project, we tested three different models to forecast sales, and the goal was to compare their performance and select the most accurate one.



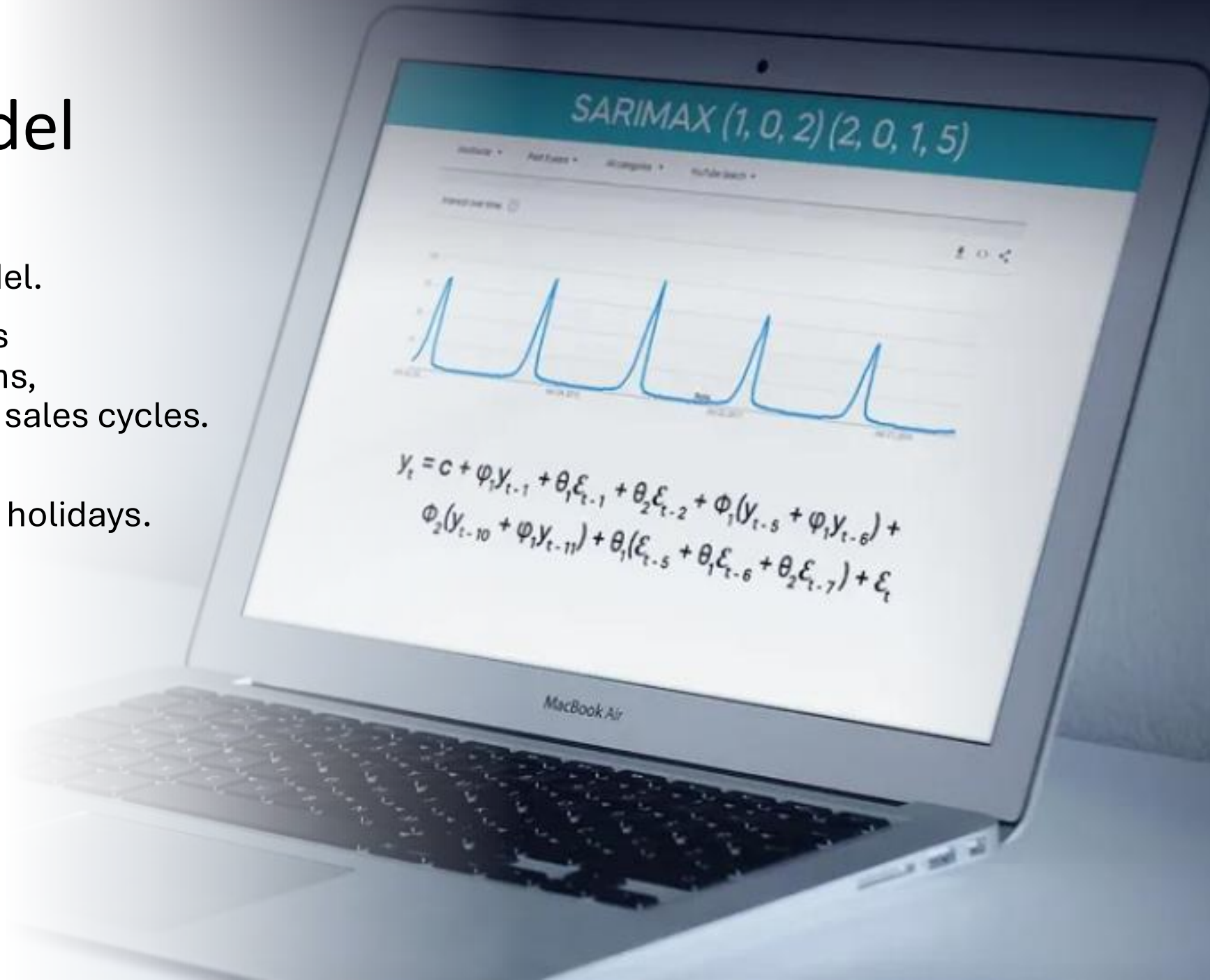
Why Multiple Models?

- We choose to test different types of models because each one captures different patterns in the data.
- Time-series models understand seasonality, while machine-learning models capture complex interactions.



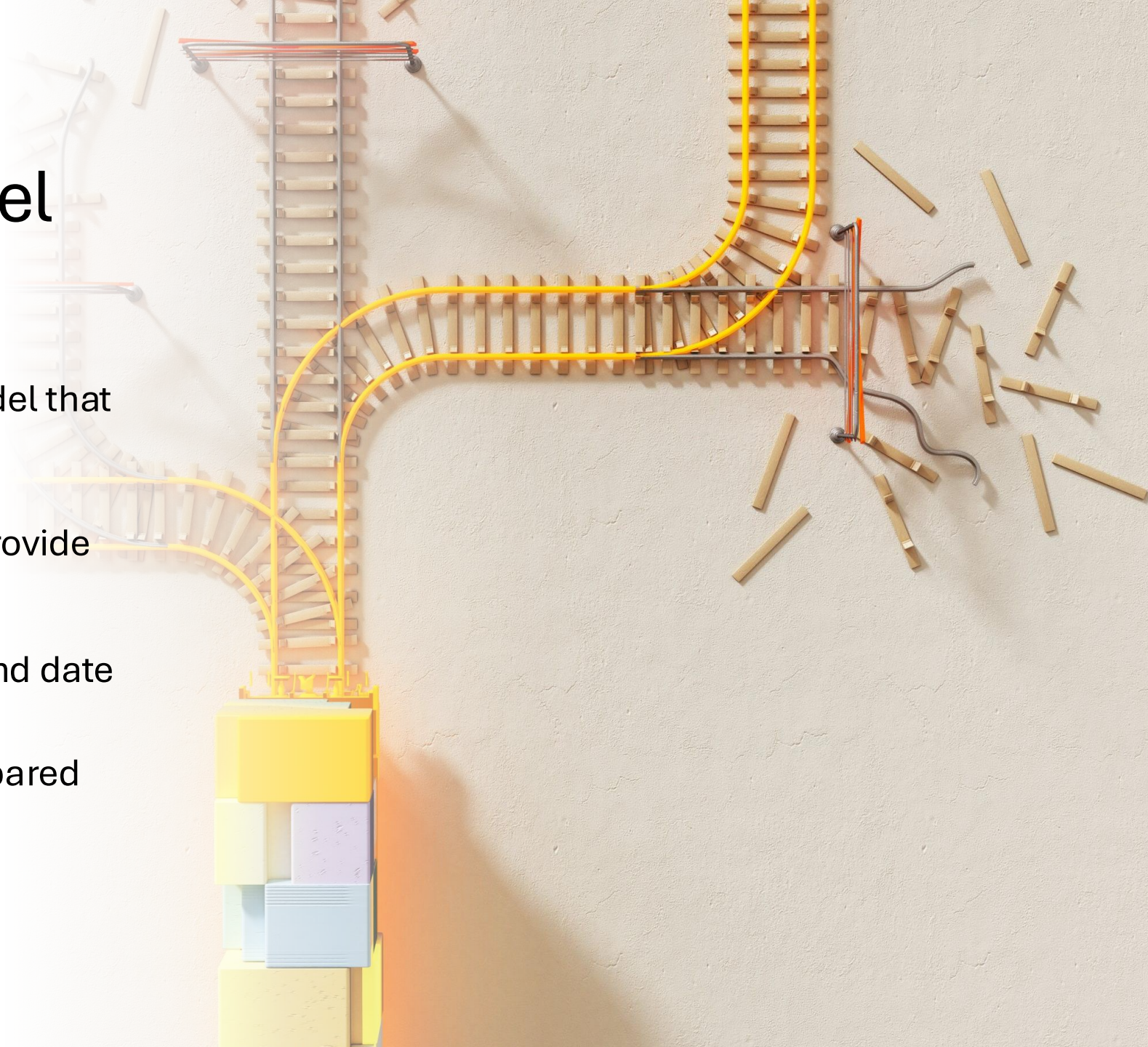
SARIMAX Model

- A classical time-series model.
- It captures long-term trends and strong seasonal patterns, such as yearly and monthly sales cycles.
- It allows us to add external factors like promotions and holidays.



XGBoost Model

- A machine-learning model that handles non-linear relationships.
- It works well when we provide engineered features like promotion duration, competition duration, and date features.
- It is slower to train compared to LightGBM.



LightGBM Model

- A faster and more optimized version of gradient boosting.
- It trains much faster than XGBoost and gave us better accuracy than XGBoost.
- LightGBM delivered the second-best performance overall.



Final Model Selection

- After comparing the results, SARIMAX achieved the lowest RMSE.
- So, we selected SARIMAX as our final model.

	SARIMA	XGBoost	LightGBoost
RMSE	867.95	1166.51	1056.42



MLOPS



What is MLflow



MLflow is an open-source platform used to manage the end-to-end machine learning lifecycle.



It helps data scientists track experiments, manage models, and deploy them consistently.

Why Do We Use MLflow?

- Machine learning experiments produce many models and parameters
- Hard to track :
 - Which model performed best
 - Which hyperparameters were used
 - Which data version was trained on

MLOps Flow Overview

End-to-end pipeline for training, tracking, packaging, and serving models

Artifacts saved locally then logged to MLflow

Pyfunc wrappers ensure standardized inference

Supports SARIMAX, and LightGBM models

Experiment Tracking (MLflow UI)

- MLflow dashboard displaying experiments
- Organizes and audits model runs
- Tracks parameters, metrics, and artifacts
- Core tool for reproducible ML workflows



MLflow UI

 3.6.0

+ New

Home

Experiments

Models

Prompts

Welcome to MLflow

Get started

**Log traces**
Trace LLM applications for debugging and monitoring.

**Run evaluation**
Iterate on quality with offline evaluations and comparisons.

**Train models**
Track experiments, parameters, and metrics throughout training.

**Register prompts**
Manage prompt updates and collaborate across teams.

Experiments

Name

Time created

Last modified

Description

Tags

rossmann

11/18/2025, 09:29:54 PM

11/30/2025, 06:41:21 AM

-

rossmann_full_retro

11/26/2025, 03:47:30 PM

11/26/2025, 03:47:30 PM

-

rossmann_full

11/26/2025, 03:45:22 PM

11/26/2025, 03:45:22 PM

-

rossmann_test_pyfunc

11/26/2025, 03:24:22 PM

11/26/2025, 03:24:22 PM

-

Default

11/08/2025, 09:58:41 PM

11/08/2025, 09:58:41 PM

-

View all

Model Comparison

+ New

- Home
- Experiments
- Models
- Prompts

fossmanth

Comparing 2 Runs from 1 Experiment

Visualizations

Parallel Coordinates Plot Scatter Plot Box Plot Contour Plot

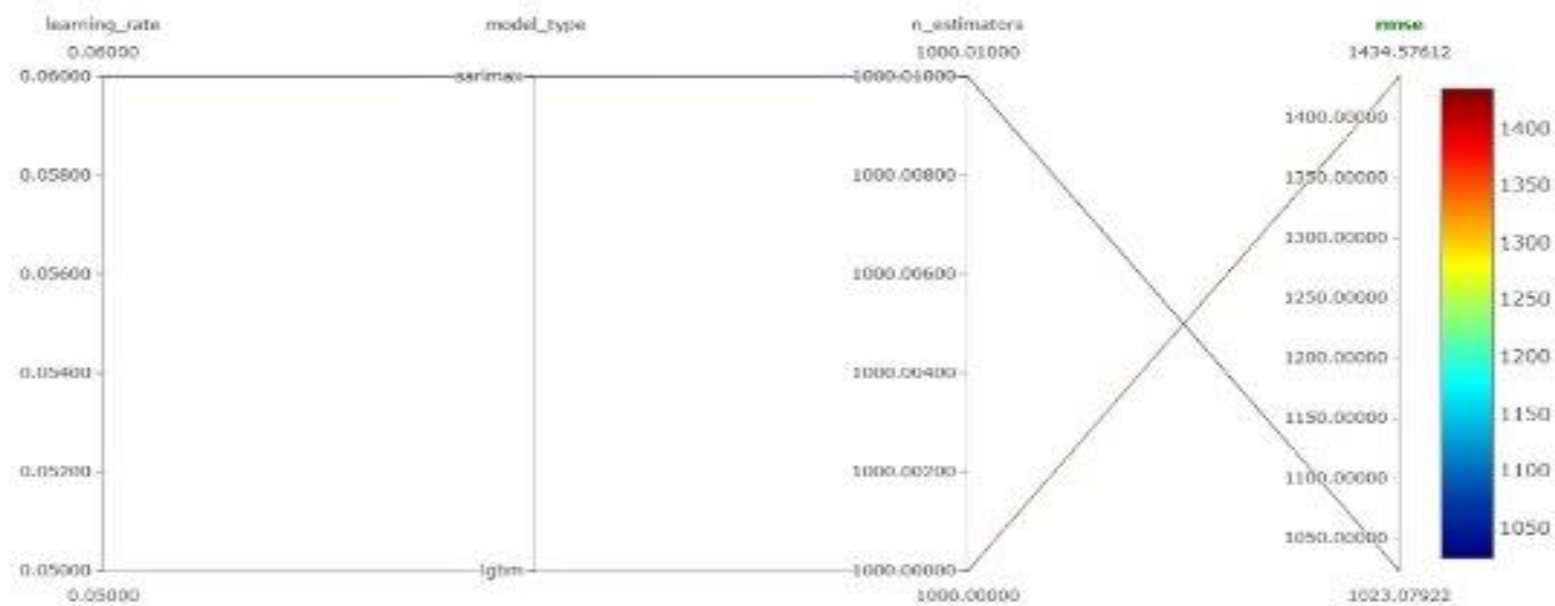
Parameters:

learning_rate X model_type X
n_estimators X

Metrics:

rmse X

Clear All



> Run details

Conclusion and Future Improvements

What is Project Deployment?

- Is the process of taking a developed model or application from a development environment and making it available for real-world use.

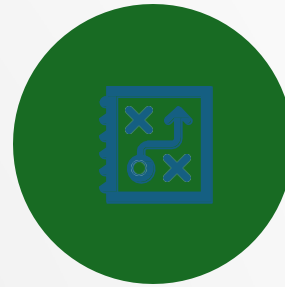


Benefits of Project Deployment



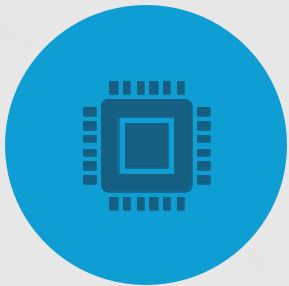
- **Real-Time Predictions**

Allows models to generate predictions on live data instead of notebooks.



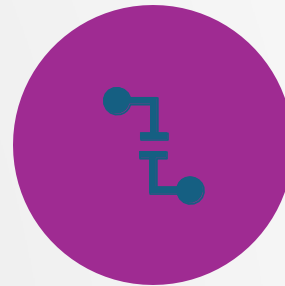
- **Business Impact**

Transforms analytical insights into actionable decisions.



- **Scalability**

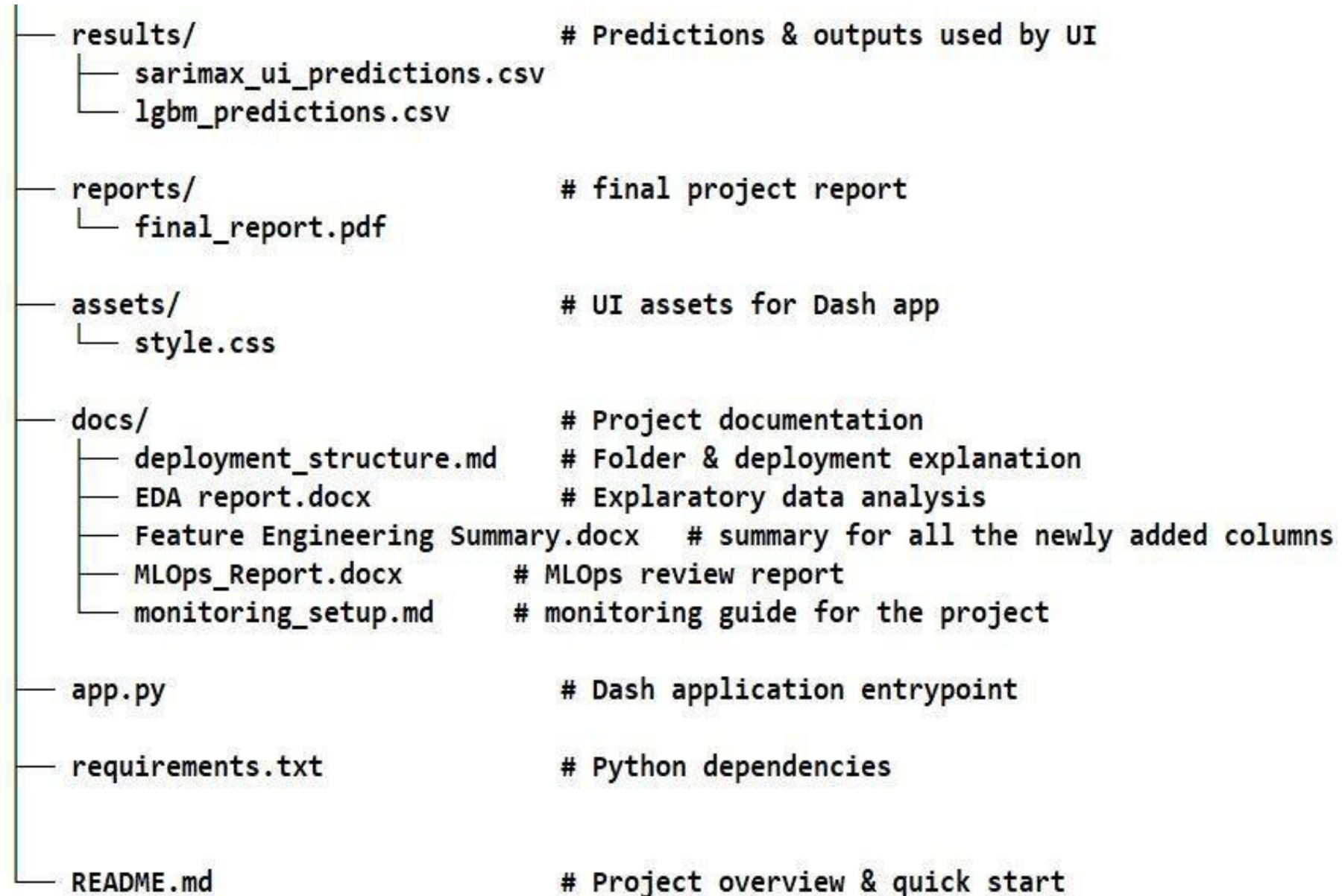
Enables predictions for thousands of users or stores simultaneously.



- **Consistency & Reliability**

Ensures the same model logic is used every time.

Project Structure



Web App

Rossmann SARIMAX Forecast

Interactive forecasting UI — upload data or use the sample test set

Version 2.0.0

☒ Use dataset/test.csv

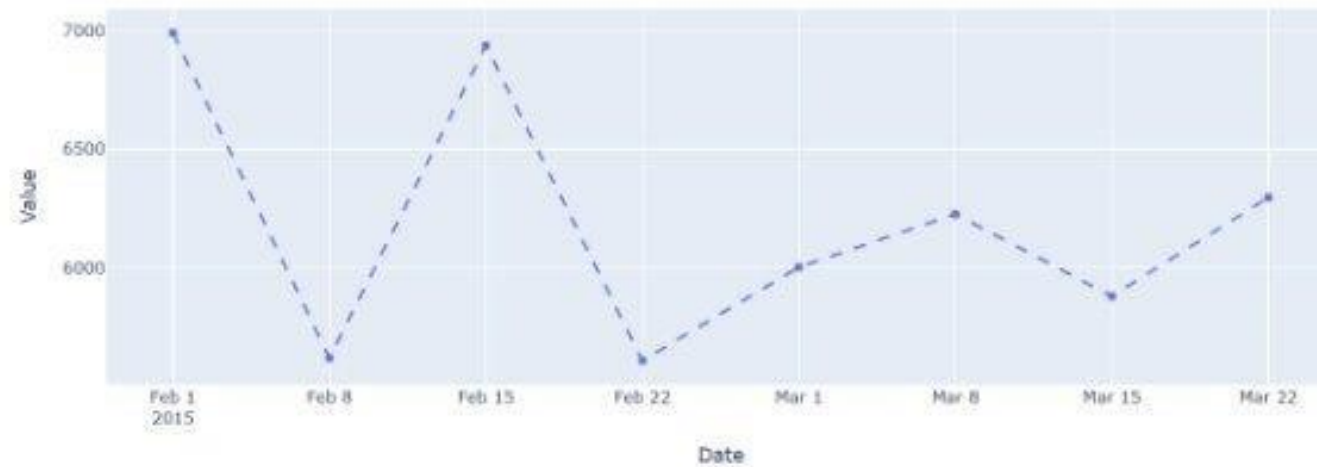
Drag & drop or select a CSV file

Forecast horizon (weeks): 8

Forecast

Forecast complete — saved to results/sarimax_ui_predictions.csv (horizon=8 weeks)

Historical and Forecasted Sales



[Errors](#) [Callbacks](#)

v3.2.0

[Dash update available - v3.3.0](#)

Server



Room for Improvements



Improve RMSE and use other metrics



Deep learning models like LSTM, GRU, and Temporal Convolutional Networks

Room for Improvements

Per-Store Modeling

Incorporate lag features, rolling windows, weather data, macroeconomic indicators, and geospatial factors.

Deploy project on the cloud

The background is a light blue gradient. It features several abstract elements: a light blue blob in the top left with a small yellow circle inside; a green blob in the top right with a blue circle and a thin purple line; a purple blob in the bottom left with a yellow circle and a thin purple line; and a yellow blob in the bottom right with a small yellow circle and a thin purple line. The text is centered in the middle of the image.

THANK YOU

FOR YOUR APPRECIATION



ANY QUESTIONS ?